

Assignment 3 Solutions

Fall 2026

$$2.2 \quad 4 \begin{bmatrix} -1 \\ 1 \\ 10 \end{bmatrix} - 5 \begin{bmatrix} -6 \\ 3 \\ 7 \end{bmatrix} + 9 \begin{bmatrix} 8 \\ 9 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \\ 40 \end{bmatrix} - \begin{bmatrix} -30 \\ 15 \\ 35 \end{bmatrix} + \begin{bmatrix} 72 \\ 81 \\ 9 \end{bmatrix} = \begin{bmatrix} -4 + 30 + 72 \\ 4 - 15 + 81 \\ 40 - 35 + 9 \end{bmatrix} = \begin{bmatrix} 98 \\ 70 \\ 14 \end{bmatrix}$$

2.9 This is not possible. Since $v_2 = -4v_1$, $v_1 \in \text{span}\{v_2\}$ and

$v_2 \in \text{span}\{v_1\}$, which implies $\text{span}\{v_1\} = \text{span}\{v_2\} = \text{span}\{v_1, v_2\}$

$$2.13 \quad \begin{bmatrix} 1 & -2 & -8 & 16 \\ -2 & 5 & 21 & -38 \\ 0 & 2 & 10 & -12 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -8 & 16 \\ -2 & 5 & 21 & -38 \\ 0 & 1 & 5 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -8 & 16 \\ 0 & 1 & 5 & -6 \\ 0 & 1 & 5 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -8 & 16 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 4 - 2x_3 \\ x_2 &= -6 - 5x_3 \\ x_3 &\text{ is free} \end{aligned}$$

$$2.18 \quad \begin{bmatrix} -2 & -1 & v_1 \\ 1 & 0 & v_2 \\ 2 & 2 & v_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & v_2 \\ -2 & -1 & v_1 \\ 2 & 2 & v_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & v_2 \\ 0 & -1 & v_1 + 2v_2 \\ 0 & 2 & v_3 - 2v_2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & v_2 \\ 0 & -1 & v_1 + 2v_2 \\ 0 & 0 & v_3 - 2v_2 + 2(v_1 + 2v_2) \end{bmatrix}$$

must be 0 for the echelon form to not have an inconsistent row.

$$v_3 - 2v_2 + 2(v_1 + 2v_2) = 2v_1 + 4v_2 - 2v_2 + v_3 = 2v_1 + 2v_2 + v_3$$

$$2v_1 + 2v_2 + v_3 = 0$$

$$\begin{aligned} \text{Check: } 2(-2) + 2(1) + 2 &= 0 \checkmark \\ 2(-1) + 2(0) + 2 &= 0 \checkmark \end{aligned}$$

Idea: The span of $\begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$ is exactly those vectors $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ for

which $x \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ has a solution. And this equation has a solution if its echelon form does not have an inconsistent row.

$$3.1 \quad \begin{bmatrix} -10 & 6 & 2 & 8 \\ 1 & 3 & 4 & 5 \\ 0 & -2 & 0 & -9 \end{bmatrix} + \begin{bmatrix} 4 \\ -5 \\ 3 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} -10 \\ 1 \\ 0 \end{bmatrix} - 5 \begin{bmatrix} 6 \\ 3 \\ -2 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 8 \\ 5 \\ -9 \end{bmatrix}$$

$$\begin{bmatrix} -40 \\ 4 \\ 0 \end{bmatrix} - \begin{bmatrix} 30 \\ 15 \\ -10 \end{bmatrix} + \begin{bmatrix} 6 \\ 12 \\ 0 \end{bmatrix} + \begin{bmatrix} 8 \\ 5 \\ -9 \end{bmatrix} = \begin{bmatrix} -40 - 30 + 6 + 8 \\ 4 - 15 + 12 + 5 \\ 0 + 10 - 9 \end{bmatrix} = \begin{bmatrix} -56 \\ 6 \\ 1 \end{bmatrix}$$

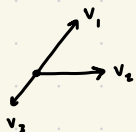
3.3 Not possible. Matrix-vector multiplication is defined only when the number of entries of the vector is the same as the number of columns of the matrix.

$$3.9 \quad \begin{bmatrix} 1 & 2 \\ -2 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \text{Full span, one pivot per row}$$

$$3.12 \quad \begin{bmatrix} 1 & 5 & -1 \\ -2 & -10 & +2 \\ 2 & -4 & 5 \\ -4 & -20 & +4 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & -1 \\ 0 & -14 & 7 \\ 0 & -20 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{Not full span}$$

2.12 True. $u = -(v_1 + v_2)$, $v_1 + v_2 + u = v_1 + v_2 + (-(v_1 + v_2)) = v_1 + v_2 - v_1 - v_2 = (v_1 - v_1) + (v_2 - v_2) = \vec{0} + \vec{0} = \vec{0}$

2.23 False.



$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$v_1 = 0v_2 - v_3 \quad v_2 \notin \text{span}\{v_1, v_3\} = \text{span}\{v_1\}$$

2.24 False. $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \quad \text{span}\{v_1, v_2\} \text{ is not a plane}$

3.18 False. $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ 1 pivot, 2 rows, but has a free variable

3.19 False. $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ but $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \notin \{v : Av = \vec{0}\}$

3.23 False. $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$

2.26 $v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ Note that every vector in $\text{span}\{v_1, v_2\}$ has the property that the first and second entry are the same

$v_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ Note that every vector in $\text{span}\{v_1, v_2, v_3\}$ has the property that the third and fourth entry are the same.

2.27 Idea: We can determine linear equations for each span, then find a solution for both equations

$$\begin{bmatrix} 1 & 0 & v_1 \\ -2 & 1 & v_2 \\ -1 & 1 & v_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & v_1 \\ 0 & 1 & v_2 - 2v_1 \\ 0 & 1 & v_3 + v_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & v_1 \\ 0 & 1 & v_2 - 2v_1 \\ 0 & 0 & v_3 + v_1 - (v_2 - 2v_1) \end{bmatrix}$$

$$v_3 + v_1 - (v_2 - 2v_1) = v_3 + v_1 - v_2 + 2v_1 = 3v_1 - v_2 + v_3$$

$$3v_1 - v_2 + v_3 = 0$$

check:

$$3(1) - 2 + (-1) = 0 \checkmark$$

$$3(0) - 1 + 1 = 0 \checkmark$$

$$\begin{bmatrix} 1 & 0 & v_1 \\ -1 & 1 & v_2 \\ -1 & 3 & v_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & v_1 \\ 0 & 1 & v_2 - v_1 \\ 0 & 3 & v_3 - v_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & v_1 \\ 0 & 1 & v_2 - v_1 \\ 0 & 0 & v_3 - v_1 - 3(v_2 - v_1) \end{bmatrix}$$

$$v_3 - v_1 - 3v_2 + 3v_1, \quad 2v_1 - 3v_2 + v_3 = 0$$

$$2v_1 - 3v_2 + v_3 = 0$$

$$v = \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}$$

$$x_1 = -2/7 x_3$$

$$x_2 = \frac{1}{7} x_3$$

x_2 is free

already in echelon form

$$v = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} & & & -1 \\ & & & 1 \\ & & & -1 \\ \dots & \dots & \dots & 1 \\ & & & -1 \\ & & & 1 \end{bmatrix}$$